Pavel Vasilyev

Senior AI Engineer - LLM Systems, APIs & Applied ML

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PROFILE

Senior AI engineer with fifteen years across applied ML, NLP, and software. I build LLM systems where routing, fallback, cost, and latency are explicit and measurable, and I design the APIs and data flows around them.

FIT FOR THE ROLE

- Multi-step LLM workflows across Claude, Gemini, OpenRouter, Ollama, and Cloudflare Workers AI: provider routing, role-specific fallback chains, retry ceilings, time and token budgets, bounded concurrency, and per-call token and latency tracing
- Classical ML and NLP depth: feature engineering, model selection and training, text classification and extraction
- Clean API and service design: authenticated HTTP APIs, credential handling, Phoenix services deployed behind Caddy and systemd
- Working with product and design: turning ambiguous requests into written specifications and explicit trade-off decisions

EXPERIENCE

Independent Software Practice - consulting and open source 2010-present

Consulting, advisory, and software practice between salaried roles.

- Built tools and technical systems against client specifications while maintaining an open-source body of 150+ projects across eight or more languages.

Research Lead - Spring Research 2020-2022

Quantitative research firm focused on financial-market modelling.

- Owned the full applied-research cycle: monitored financial markets, analysed data, developed new signal representations and transformations, extracted ML features, selected approaches, and built and trained models.

Earlier Data Science, NLP & Software Roles - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

Roles in Tel Aviv, Hong Kong, and Prague.

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer-vision pipelines; and corpus-linguistics infrastructure.

SELECTED WORK (PRIVATE REPOSITORIES, CODE ON REQUEST)

pi-harness - Eval harness for a coding agent, scoring pass/fail, cost, and wall time per run. A/B of a structural gate: spec-first 0/3 to 3/3 at +26% cost and +80% wall time (n=3, one model).

inference_harness - Elixir LLM harness for multi-provider routing, role-specific fallback, token/latency tracing, and bounded concurrent jobs.

gemini-gate - Elixir/Phoenix credential service with authenticated retrieval, Gemini API proxying, and Caddy/systemd deployment.

BACKGROUND

Education: MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Languages: English, Hebrew, Russian; working knowledge of German and French.